

Blatt DTL

1)

Nr.	x_1 Alter	x_2 Einkommen	x_3 Bildung	Kandidat
1	≥ 35	hoch	Abitur	O
2	< 35	niedrig	Master	O
3	≥ 35	hoch	Bachelor	M
4	≥ 35	niedrig	Abitur	M
5	≥ 35	hoch	Master	O
6	< 35	hoch	Bachelor	O
7	< 35	niedrig	Abitur	M

$$S_1 = 4 \quad S_2 = 0.7$$

AL3

*

- 1 /01/
- 2 /02/
- 3 /02 M1/
- 4 /02 M2/

$$x_1(*, /A1/)$$

$$5 \quad x_1(*, /M1O1/)$$

$$6 \quad x_1(/01/, /M1O1/)$$

$$7 \quad x_1(/01M/, /M1O1/)$$

$$P(O) = 2/4 = 0.5$$

$$P(M) = 2/4 = 0.5$$

$$1 \quad x_1(/01M1/, /M1O2/)$$

$$2 \quad x_1(/02M1/, /M1O2/)$$

$$3 \quad x_1(/02M1/, /M2O2/)$$

$$x_1(/02M1/, x_2(*, /A1/))$$

$$4 \quad x_1(/02M1/, x_2(/M1/, /M1/))$$

$$5 \quad x_1(/02M1/, x_2(/M1/, /M2/))$$

$$6 \quad x_1(/03M1/, x_2(/M1/, /M2/))$$

$$P(O) = 2/4 = 0.5$$

$$P(M) = 2/4 = 0.5$$

$$P(O) = 3/4 = 0.75$$

$$P(M) = 1/4 = 0.25$$

$$x_2(O, x_2(/M1/, /M2/))$$

$$1 \quad x_2(O, x_2(/M1/, /M2O1/))$$

$$3 \quad x_2(O, x_2(/M1/, /M3O1/))$$

$$x_2(O, x_2(/M1/, M))$$

$$4 \quad x_2(O, x_2(/M2/, M))$$

$$4 \quad x_2(O, x_2(/M3/, M))$$

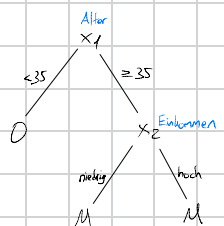
$$4 \quad x_2(O, x_2(/M4/, M))$$

$$x_2(O, x_2(M, M))$$

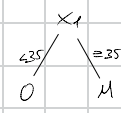
M dominiert

M dominiert

Baum:



bzw →



1D3

Nr.	x_1 Alter	x_2 Einkommen	x_3 Bildung	Kandidat
1	≥ 35	hoch	Abitur	0
2	< 35	niedrig	Master	0
3	≥ 35	hoch	Bachelor	M
4	≥ 35	niedrig	Abitur	M
5	≥ 35	hoch	Master	0
6	< 35	hoch	Bachelor	0
7	< 35	niedrig	Abitur	M

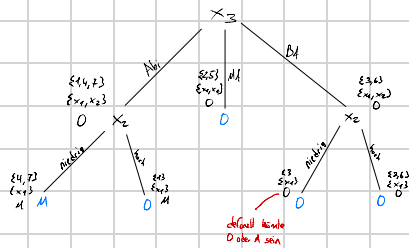
$$S_1 = 4 \quad S_2 = 0,7$$

$$x_1 = \geq 35 \quad \{1,3,4,5\} \rightarrow p(x_1) = 4/7$$

$$x_1 = < 35 \quad \{2,6,7\} \rightarrow p(x_1) = 3/7$$

$$\begin{aligned} R(S, A) &= \frac{4}{7} \cdot (H(\{1,3,4,5\})) + \frac{3}{7} \cdot (H(\{2,6,7\})) \\ &= \frac{4}{7} \cdot \left(\frac{3}{4} \cdot \log_2 \frac{3}{4} - \frac{3}{4} \cdot \log_2 \frac{3}{4} \right) + \\ &\quad \frac{3}{7} \cdot \left(-\frac{2}{3} \cdot \log_2 \frac{2}{3} - \frac{1}{3} \cdot \log_2 \frac{1}{3} \right) \\ &= \frac{4}{7} + 0,39 = 0,96 \end{aligned}$$

$\{1,2,3,4,5,6,7\} \quad \{x_1, x_2, x_3\} \quad 0$



$$x_2 = \text{hoch} \quad \{1,3,5,6\} \rightarrow \frac{4}{7}$$

$$x_2 = \text{niedrig} \quad \{2,4,7\} \rightarrow \frac{3}{7}$$

$$\begin{aligned} R(S, A) &= \frac{4}{7} \cdot (H(\{1,3,5,6\})) + \frac{3}{7} \cdot (H(\{2,4,7\})) \\ &= \frac{4}{7} \cdot \left(-\frac{3}{4} \cdot \log_2 \frac{3}{4} - \frac{1}{4} \cdot \log_2 \frac{1}{4} \right) + \\ &\quad \frac{3}{7} \cdot \left(-\frac{1}{3} \cdot \log_2 \frac{1}{3} - \frac{2}{3} \cdot \log_2 \frac{2}{3} \right) \\ &= 0,46 + 0,39 = 0,85 \end{aligned}$$

$$x_3 = \text{Abitur} \quad \{1,4,7\} \rightarrow \frac{3}{7}$$

$$x_3 = \text{Master} \quad \{2,5\} \rightarrow \frac{2}{7}$$

$$x_3 = \text{Bachelor} \quad \{3,6\} \rightarrow \frac{2}{7}$$

$$\begin{aligned} R(S, A) &= \frac{3}{7} \cdot (H(\{1,4,7\})) + \frac{2}{7} \cdot (H(\{2,5\})) + \frac{2}{7} \cdot (H(\{3,6\})) \\ &= \frac{3}{7} \cdot \left(-\frac{1}{3} \cdot \log_2 \frac{1}{3} - \frac{2}{3} \cdot \log_2 \frac{2}{3} \right) + \\ &\quad \frac{2}{7} \cdot \left(-\frac{1}{2} \cdot \log_2 \frac{1}{2} - \frac{1}{2} \cdot \log_2 \frac{1}{2} \right) \\ &= 0,39 + 0 + 0,29 = 0,68 \end{aligned}$$

2)

$x_3(x_1(x_2(C, A), x_1(B, A)), x_1(x_2(C, B), A))$ → transformation
 $x_3(x_1(x_2(C, B), x_2(A, A)), x_1(x_2(C, B), A))$ → Bed. irrelevant
 $x_3(x_1(x_2(C, B), A)), x_1(x_2(C, B), A))$ → transformation
 $x_4(x_3(x_2(C, B), x_2(C, B)), x_3(A, A))$ → transformation
 $x_4(x_2(x_3(C, C), x_2(B, B), x_3(A, A))$ → transformation
 $x_4(x_2(C, B), x_3(A, A))$ → Bed. irrelevant
 $x_4(x_2(C, B), A)$ → Bed. irrelevant

